Encryption

CONFIDENTIAL

RC4-128 · permission flags · open and owner passwords

How It Works

PDF encryption (RC4-128, algorithm V=2 R=3) is applied as a post-processing pass after document assembly. The engine derives a 128-bit file encryption key from the password and permission flags, then encrypts all string literals and stream bytes in-place. Object offsets stay valid because RC4 is length-preserving.

API

Call setEncryption before renderPdf — the PDF bytes are then returned already encrypted.

Permissions only (no open password)
engine.setEncryption("", "s3cr3t", { print: false, copy: false })
File opens without prompting; cooperative viewers enforce the flags.

With open password
engine.setEncryption("password", "owner", { copy: false })
Viewers prompt for the user password before displaying content.

Permission Flags

Flag	Effect when disabled
print	Printing disabled in viewer
copy	Text and graphic extraction blocked
modify	Content modification blocked
annotate	Adding or modifying annotations blocked
fill_forms	Form field filling disabled
assemble	Page insertion, deletion, rotation blocked
print_hq	High-quality printing degraded to low resolution

Note on strength
Permission flags are advisory — cooperative viewers honour them, but tools such as qpdf can strip them in seconds when the open password is empty. They are a deterrent, not a lock. For strong access control, pair with an open password.